

Diagnosing Causation Claims and Biased Study Designs

Answer Key

Precalculus / Statistics

Quick Answers

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|---|---|
| 1. (a) the share of all sales = $\frac{1}{625}$, — | 6. (a) the absolute risk difference = $\frac{1}{1000}$, — |
| 2. (a) the proportion of respondents = $\frac{7}{8}$, — | 7. (a) the proportion that thrived when treated = $\frac{3}{4}$, — |
| 3. (a) the value of r squared = 0.81, — | 8. (a) the four-month total = 780, (b) the |
| 4. (a) the proportion in the sample = $\frac{3}{4}$, — | busiest month's admissions = 260, — |
| 5. (a) mean of the course group = 82, (b) mean of the no-course group = 74, — | |
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What is verified

10 of 18 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus / Statistics

HSS-IC.A – *problems 2, 4*

HSS-IC.B – *problems 1, 5, 6, 7*

HSS-ID.C – *problems 3, 8*

Difficulty 1–5 of 5 across 18 tagged checks.

Common wrong answers

- If they got 1: quoted the doubling as the share, reporting a 100 percent figure instead of the fraction of total sales*
 - If they got 0.07: divided by the whole membership instead of by the number who actually responded*
 - If they got 0.5: reported the relative reduction as if it were the absolute change in risk*
 - If they got 0.375: used all 400 plants as the denominator instead of the 200 that were treated*
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- “Sales doubled — up 100%!” 4 units last month, 8 this month, out of 5000 units sold.
(a) share of total sales. (b) why is the headline misleading?

Solution (a): $\frac{8}{5000}$, and dividing both parts by 8 gives $\frac{1}{625}$.

$\frac{1}{625}$

Solution (b) — instructor-judged. The doubling is real, but the base is tiny: the flavour went from 0.08% to 0.16% of sales, four extra units out of five thousand. A percentage

change computed on a very small starting number is large almost automatically, so quoting it without the base figure gives a false impression of importance. Full credit requires naming the small base as the reason.

2. 3000 members, 240 replied to a website survey, 210 liked the logo. (a) proportion of respondents. (b) why not a claim about all members?

Solution (a): $\frac{210}{240}$, dividing both parts by 30.

$\frac{7}{8}$

Solution (b) — instructor-judged. The 240 chose to answer — a voluntary response sample. Members with strong feelings are much likelier to reply than the 2760 who did not, so the respondents are not a random slice of the membership and the seven-eighths figure describes only them. Full credit requires naming voluntary response, self-selection or nonresponse bias.

3. Ice cream sales and pool rescues, $r = 0.90$. (a) r^2 . (b) name the flaw.

Solution (a): $r^2 = 0.90^2$.

0.81

Solution (b) — instructor-judged. Hot weather is a common cause: it independently raises ice cream sales and the number of people swimming, and therefore the number of rescues. Neither variable acts on the other, so however strong the correlation, it is not evidence of causation. Full credit requires naming a plausible common cause, not merely saying "correlation is not causation".

4. 60 people leaving a gym, 45 say they exercise daily. (a) sample proportion. (b) name the bias and its direction.

Solution (a): $\frac{45}{60}$, dividing both parts by 15.

$\frac{3}{4}$

Solution (b) — instructor-judged. This is a convenience sample taken at the one place where daily exercisers congregate; being there is itself evidence of exercising. People who never exercise had almost no chance of being asked, so the estimate is biased *upward* and cannot be reported for the town. Full credit requires naming the convenience or under-coverage bias and the upward direction.

5. Course group 78, 80, 82, 84, 86; no-course group 70, 72, 74, 76, 78. (a) and (b) the means. (c) does the course work?

Solution (a): $\frac{78 + 80 + 82 + 84 + 86}{5} = \frac{410}{5}$.

82

Solution (b): $\frac{70 + 72 + 74 + 76 + 78}{5} = \frac{370}{5}$.

74

Solution (c) — instructor-judged. The arithmetic gap is 8 points, but the students *chose* whether to take the course. The groups may already have differed in motivation, prior achievement or family resources, and any of those confounding variables explains the gap just as well as the course does. Only an experiment — assigning students to course or no course at random — balances the unknown variables and would support the causal claim. Full credit requires naming self-selection or confounding and proposing random assignment.

6. Risk falls from 2 per 1000 to 1 per 1000, advertised as a 50% cut. (a) absolute change. (b) why misleading?

Solution (a): $\frac{2}{1000} - \frac{1}{1000}$.

$\frac{1}{1000}$

Solution (b) — instructor-judged. Both figures are correct descriptions of the same result: the *relative* reduction is 50%, the *absolute* reduction is one case per thousand people. The headline quotes the larger-sounding relative number and omits the base rate, so a reader imagines a benefit far bigger than one person in a thousand. Full credit requires both figures named and the point that the relative figure hides the base rate.

7. Fertilizer on the sunny half: 150 of 200 treated thrived, 100 of 200 untreated thrived. (a) proportion thriving when treated. (b) flaw and fix.

Solution (a): $\frac{150}{200}$, dividing both parts by 50.

$\frac{3}{4}$

Solution (b) — instructor-judged. Every treated plant was also a sunny plant, so sunlight and fertilizer are completely confounded: the higher survival could be caused by either, and the data cannot separate them. The fix is to assign the treatment at random across the whole field, or — better, since sun exposure is known to matter — to *block* by half and randomize within each half, so both treatments appear in sun and in shade. Full credit requires naming the confounding and proposing randomization or blocking.

8. **Synthesis.** Admissions Nov 120, Dec 210, Jan 260, Feb 190; flu shots run over the same months. (a) total. (b) busiest month. (c) diagnose the claim.

Solution (a): $120 + 210 + 260 + 190$.

780

Solution (b): The largest of the four monthly figures is January's.

260

Solution (c) — instructor-judged. The two series rise together because winter drives both: cold weather fills hospitals and is also when vaccination campaigns are scheduled. The season is a common cause, so shared timing is no evidence at all of a causal link. The direction could even run the other way, since people who feel at risk are the ones who both seek a shot and seek care. Settling the question requires a randomized trial: assign people at random to receive the vaccine or not and compare admission rates between the groups. Full credit requires the common-cause diagnosis and a randomized comparison as the remedy; noting the possible reverse direction is a strong bonus.